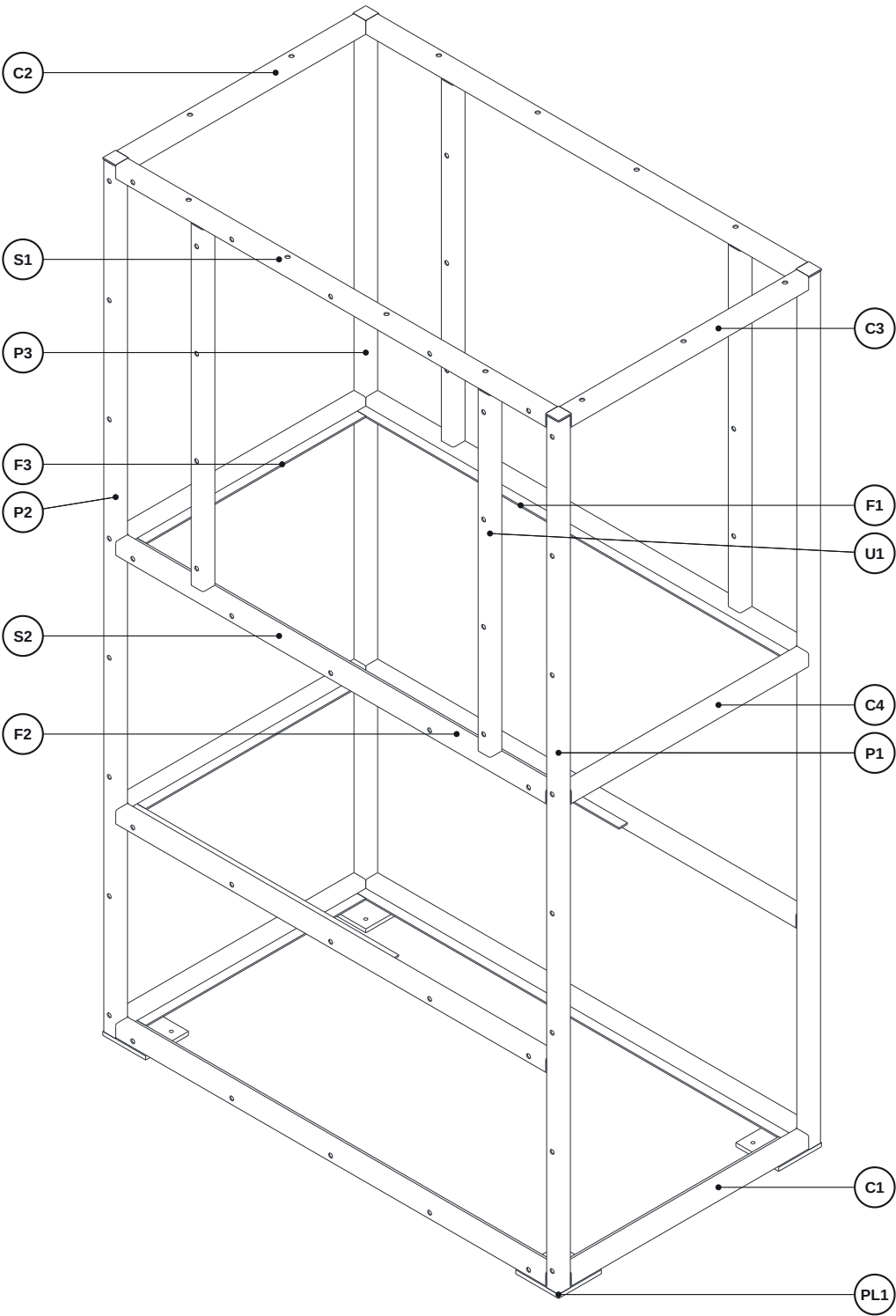
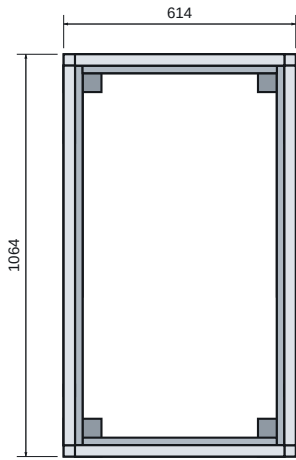
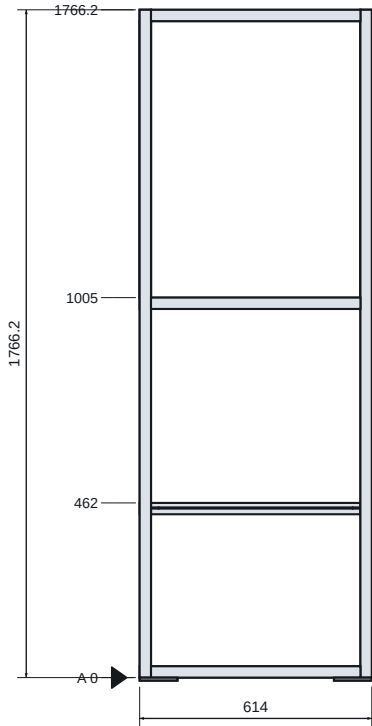




ISOMETRIC - FRONT LEFT (NOT TO SCALE)

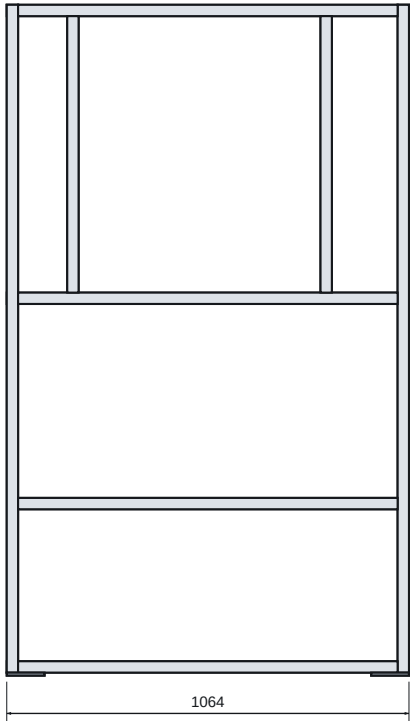


PLAN



FRONT

Datum C = left face



RIGHT SIDE

Datum B = front face (left in this view)

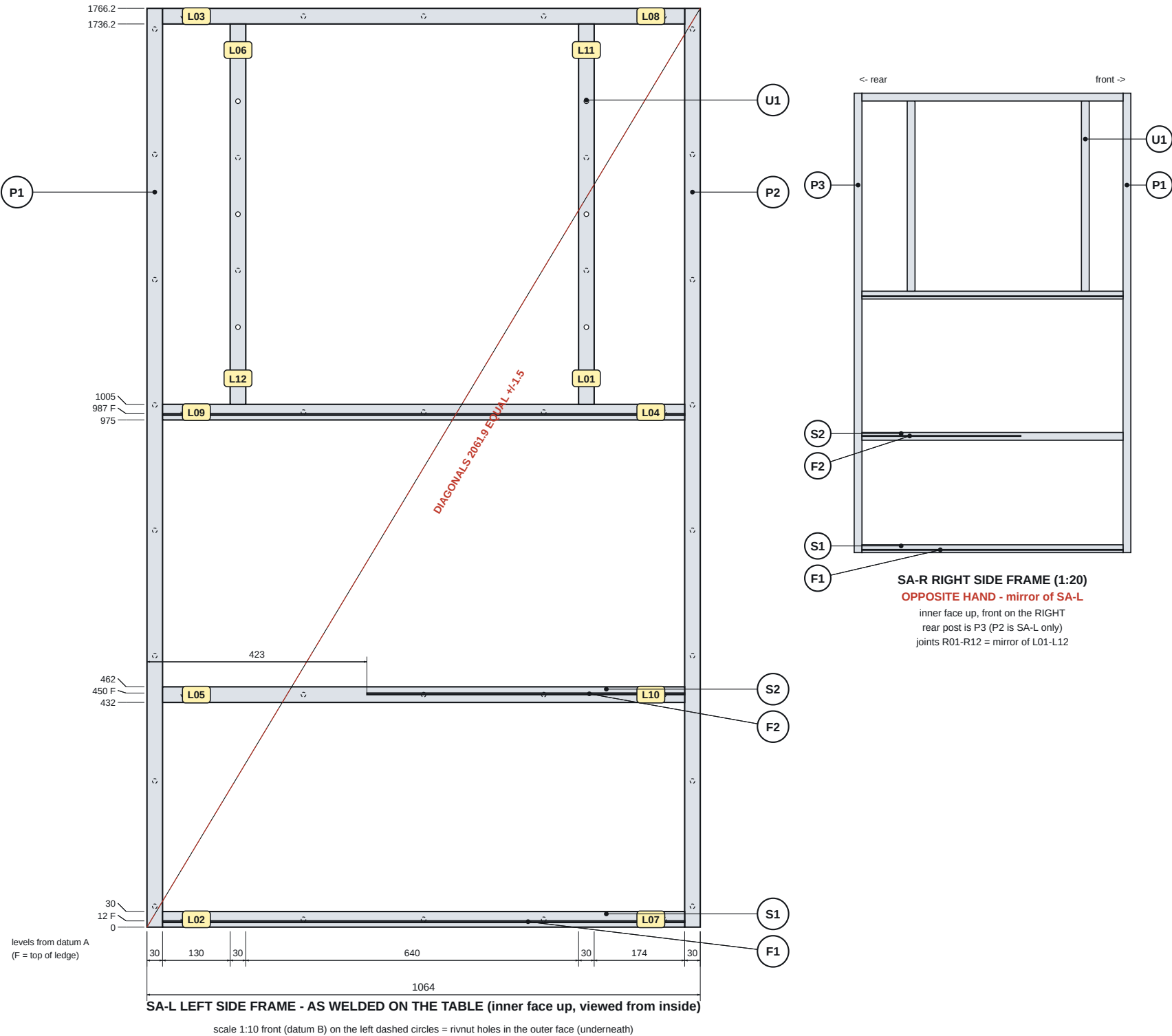
GENERAL NOTES

- 30x30x2.0 SHS C350L0 to AS/NZS 1163 (black or pre-primed; DuraGal OK - grind zinc 25 mm each side of welds + extract fumes).
Flat bar and plate grade 250 (AS/NZS 3678 / 3679.1).
- Datums (all sheets): A underside of base rails, B front face, C left face. Dimensions mm, from datums unless noted.
- GMAW, AS/NZS 1554.1 category GP. Fillets 3 mm (on 2 mm wall). SHS joints welded all round (details A/B, sheet 4).
- Faces marked * receive panels, doors or pads: grind welds flush. Build SA-L / SA-R flat (sheet 2), then box (sheet 3).
- Holes D9.0 for M6 steel rivnuts (fit after coating); drill per sheet 6 before welding. Pads PL1: 4x M8 tapped.
- Tolerances: see sheet 4. Diagonals are the key checks.
- HOLD POINT H1: mid rails (S2 mid, C3 mid rear) and ledges F2/F3 partition sit on the AC grille split M3 (462 above A).
- Tape-check M3, re-run tools/make_weld_pack.py, then weld.
- RS1 rail spacers are NOT welded: bolt to U1 after coating, pack 0-4 mm to set the 19in rails (465.1 hole centres).
- Finish: blast + zinc primer + powder coat (sheet 4).
- Weldment 43 kg (+ spacers 4.9 kg); 152 rivnuts; 38 joints.

MARK LIST (full cut list sheet 5)

MARK	QTY	DESCRIPTION	CUT / SIZE
P1	2	Post	1763.2
P2	1	Post	1763.2
P3	1	Post	1763.2
S1	4	Side rail	1004.0
S2	4	Side rail	1004.0
C1	1	Cross rail	554.0
C2	2	Cross rail	554.0
C3	3	Cross rail	554.0
C4	1	Cross rail	554.0
U1	4	Rack upright	731.2
RS1	4	Rail spacer LOOSE	731.2
F1	4	Ledge (flat bar)	1004.0
F2	2	Ledge (flat bar)	611.0
F3	4	Ledge (flat bar)	514.0
PL1	4	Castor pad, 4x M8 tapped	PL 100x100x8
PL2	4	Post top cap	PL 30x30x3

PROJECT	SRA-16 Silent AC Rack - welded 30x30x2.0 frame
TITLE	Welded frame - general arrangement + mark list
DOC NO.	CP-SRA16-FRM-001 SHEET 1 OF 6 REV A
SCALE	1:20 (iso NTS) @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-09-27 DRAWN Claude (AI) for Carbon Project
STATUS	FOR FABRICATION after hold point H1 (AC grille split M3)



SIDE FRAMES - FIT-UP AND WELD (make SA-L, then SA-R opposite hand)

- 1 Drill all members first (sheet 6). Posts outer-face holes DOWN.
- 2 Lay P1 + P2 flat, inner faces up, against a straight stop. Outside of posts 1064 apart (datum B to rear face).
- 3 Fit S1 (base, top) and S2 (mid, shelf) between the posts: S1 base flush with the post bottoms (datum A), levels per ordinates.
- 4 Stand U1 on the shelf rail at 160 and 830 from datum B.
- 5 Square each joint, clamp. Diagonals equal within 1.5, posts parallel within 1. Tack 2 per joint on the up face; re-check.
- 6 Weld in joint order L01 > L12 (balanced). Up-face seams + the short side fillets (3F uphill) - let each joint cool.
- 7 Turn over. Weld the outer-face seams in the same order, then grind flush (* faces carry the side panel). Re-check diagonals.
- 8 Inner face up again: ledges, underside stitches only (det. C):
F1 floor, top 18 below S1 top; F2 partition, top 12 below S2 (mid) top, from 423 to rear post; F1 shelf, 18 below S2 top.
Gauge: an offcut of the actual ply on the rail top.
- 9 Flat within 2 over the frame (straight-edge the outer face).

JOINTS SA-L (SA-R: R-numbers, mirror)

JOINT	MEMBER END	WELDED TO	WELD (* = external face: grind flush)
L01	U1 upright rear left	S2 Shelf left	fillet 3: front+rear; flush: outer*, inner
L02	S1 Base left	P1 post FL	fillet 3: top; flush: outer*, inner, bottom*
L03	S1 Top left	P1 post FL	fillet 3: bottom; flush: outer*, inner; top: with cap
L04	S2 Shelf left	P2 post RL	fillet 3: bottom+top; flush: outer*, inner
L05	S2 Mid left	P1 post FL	fillet 3: bottom+top; flush: outer*, inner
L06	U1 upright front left	S1 Top left	fillet 3: front+rear; flush: outer*, inner
L07	S1 Base left	P2 post RL	fillet 3: top; flush: outer*, inner, bottom*
L08	S1 Top left	P2 post RL	fillet 3: bottom; flush: outer*, inner; top: with cap
L09	S2 Shelf left	P1 post FL	fillet 3: bottom+top; flush: outer*, inner
L10	S2 Mid left	P2 post RL	fillet 3: bottom+top; flush: outer*, inner
L11	U1 upright rear left	S1 Top left	fillet 3: front+rear; flush: outer*, inner
L12	U1 upright front left	S2 Shelf left	fillet 3: front+rear; flush: outer*, inner

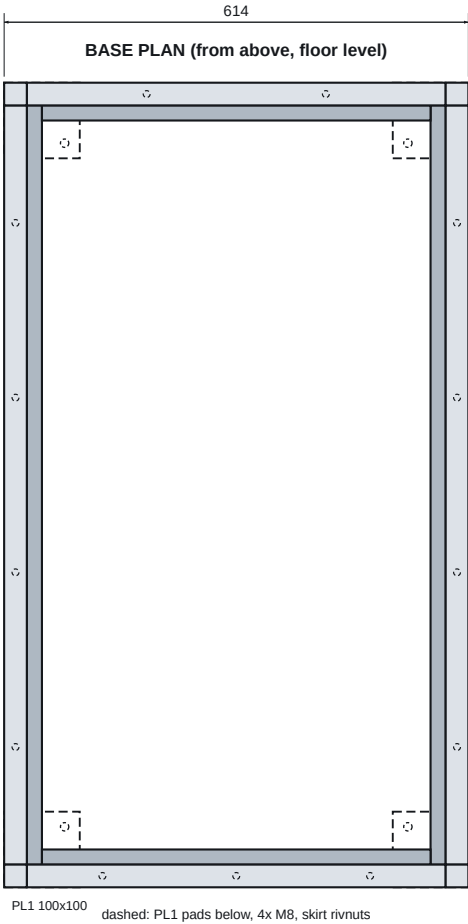
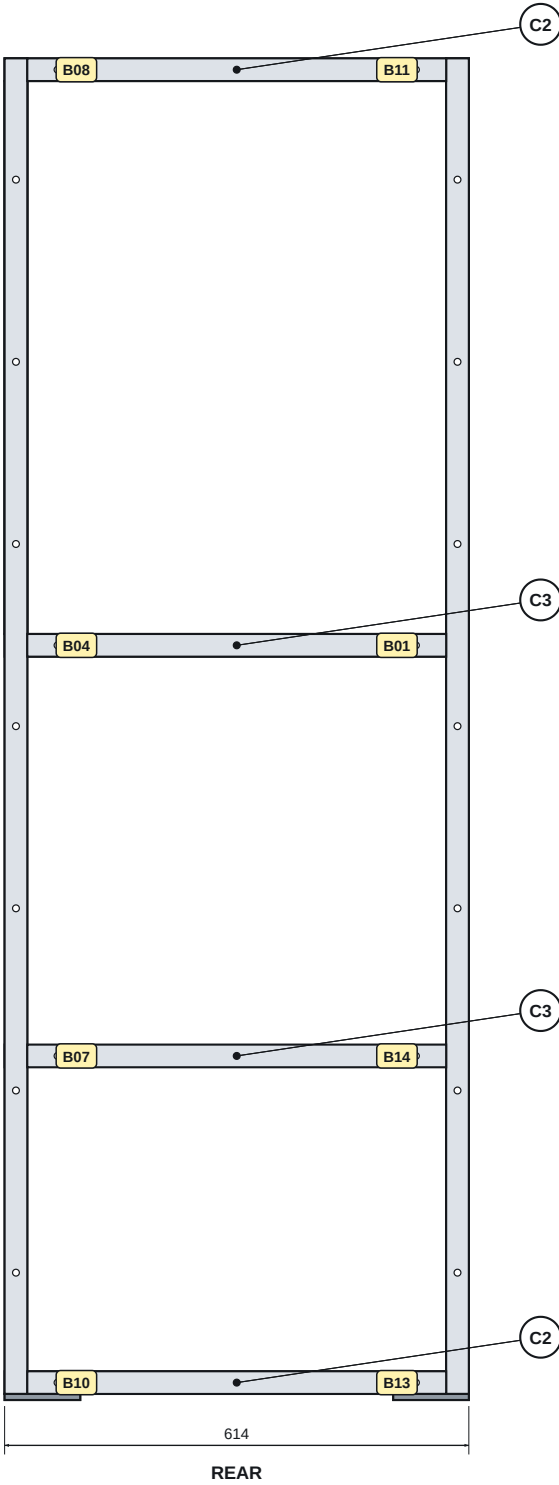
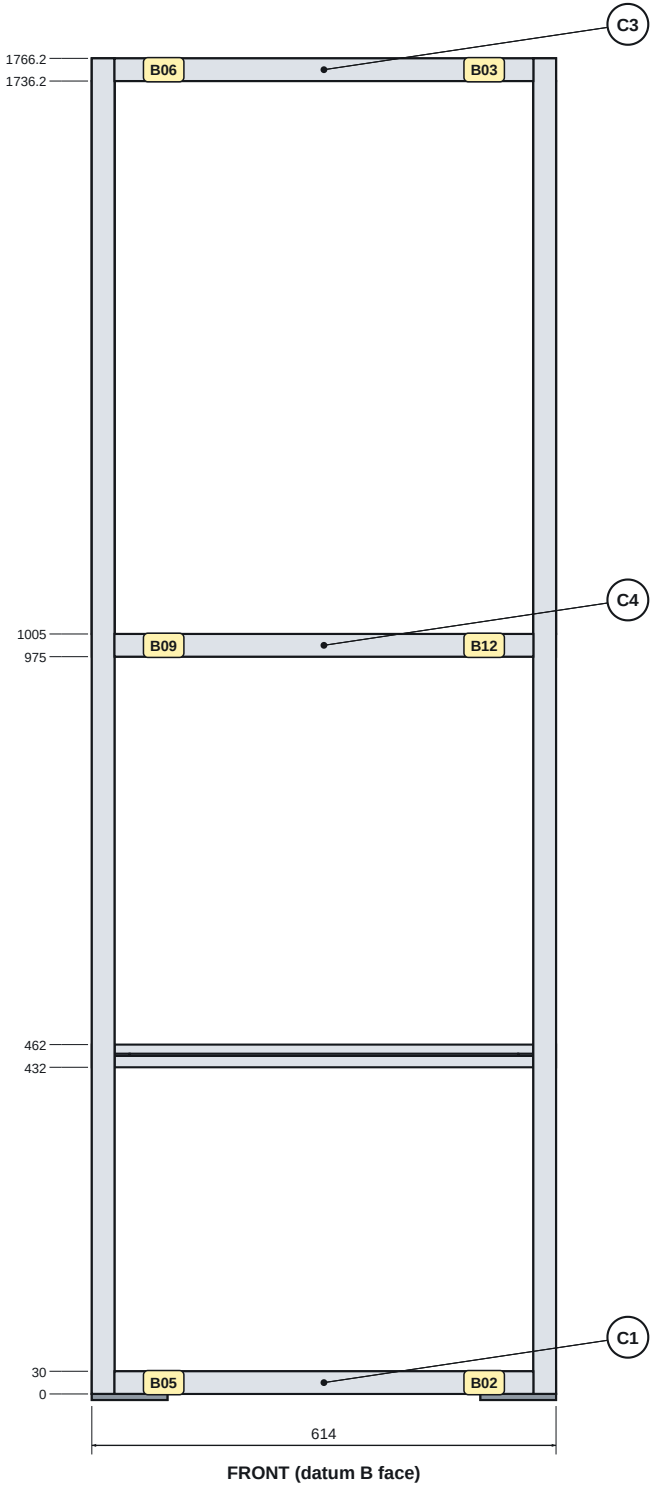
LEDGES ON THE SIDE FRAMES (det. C, underside stitches only)

WELD	FRAME	LEDGE	ON RAIL	TOP BELOW RAIL	STITCHES (@ 150 c/c)
K01	SA-L	F1 floor left	S1 Base left	18	8 x 25 + returns
K02	SA-R	F1 floor right	S1 Base right	18	8 x 25 + returns
K05	SA-L	F2 partition left	S2 Mid left	12	5 x 25 + returns
K06	SA-R	F2 partition right	S2 Mid right	12	5 x 25 + returns
K08	SA-L	F1 shelf left	S2 Shelf left	18	8 x 25 + returns
K09	SA-R	F1 shelf right	S2 Shelf right	18	8 x 25 + returns

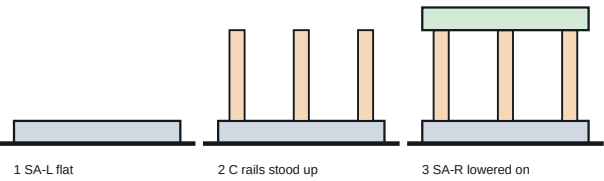
RECORD (measure tacked, then after welding)

FRAME	STAGE	DIAG 1	DIAG 2	LENGTH 1064	FLAT <= 2	INITIALS
SA-L	tacked					
SA-L	welded					
SA-R	tacked					
SA-R	welded					

PROJECT	SRA-16 Silent AC Rack - welded 30x30x2.0 frame
TITLE	Side frames SA-L / SA-R - fit-up, joints, ledges
DOC NO.	CP-SRA16-FRM-001 SHEET 2 OF 6 REV A
SCALE	1:10 / 1:20 @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-09-27 DRAWN Claude (AI) for Carbon Project
STATUS	FOR FABRICATION after hold point H1 (AC grille split M3)



BOX ASSEMBLY (on its side)



- 4 Check the four faces and body diagonals (table), twist <= 2.
- 5 Weld B01 > B14 in order, rolling the box so every joint is flat or horizontal (never overhead). Cool between joints.
- 6 Invert on the flat table: clamp PL1 pads to the table so all four are coplanar, grind bottom seams flush, weld PW1-PW4.
- 7 Weld PL2 caps T1-T4, grind all external faces flush.

BOX JOINTS (cross rails C into posts)

JOINT	RAIL END	POST	WELD (* grind flush)
B01	C3 Shelf rear	P2 RL	fillet 3: bottom+top; flush: inner, rear*
B02	C1 Base front	P1 FR	fillet 3: top; flush: front*, inner, bottom*
B03	C3 Top front	P1 FR	fillet 3: bottom; flush: front*, inner; top: with cap
B04	C3 Shelf rear	P3 RR	fillet 3: bottom+top; flush: inner, rear*
B05	C1 Base front	P1 FL	fillet 3: top; flush: front*, inner, bottom*
B06	C3 Top front	P1 FL	fillet 3: bottom; flush: front*, inner; top: with cap
B07	C3 Mid rear	P3 RR	fillet 3: bottom+top; flush: inner, rear*
B08	C2 Top rear	P3 RR	fillet 3: bottom; flush: inner, rear*; top: with cap
B09	C4 Shelf front	P1 FL	fillet 3: bottom+top; flush: front*, inner
B10	C2 Base rear	P3 RR	fillet 3: top; flush: inner, rear*, bottom*
B11	C2 Top rear	P2 RL	fillet 3: bottom; flush: inner, rear*; top: with cap
B12	C4 Shelf front	P1 FR	fillet 3: bottom+top; flush: front*, inner
B13	C2 Base rear	P2 RL	fillet 3: top; flush: inner, rear*, bottom*
B14	C3 Mid rear	P2 RL	fillet 3: bottom+top; flush: inner, rear*

CHECK BEFORE WELDING (tacked) AND AFTER

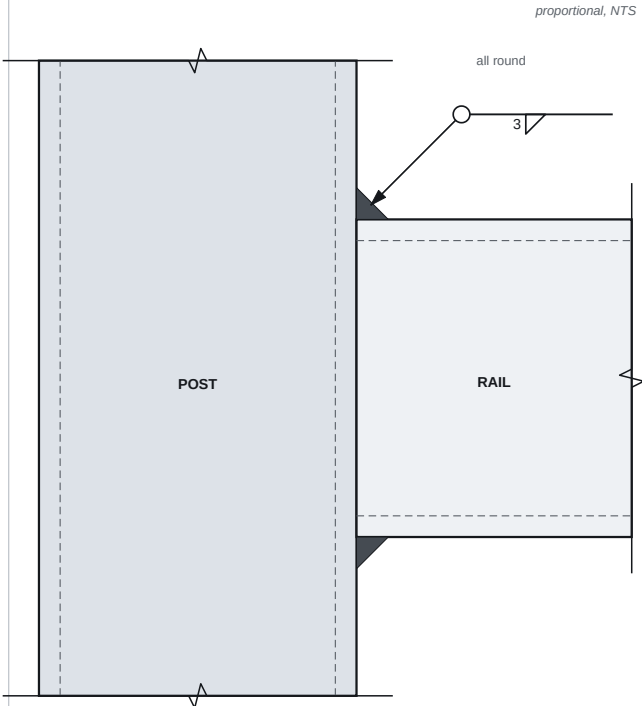
CHECK	TARGET	TOL
Front / rear face diagonals	1869.9	+/-2
Left / right face diagonals	2061.9	+/-2
Top / base diagonals	1228.5	+/-1.5
Body diagonals (4, corner-corner)	2151.4	+/-3
Outside W x D	614 x 1064	+/-1.5
Height (A to top)	1766.2	+/-1.5
Pads coplanar (rock test on table)	0	<=1
Rack uprights front-rear (19in rails)	700	+/-1

PADS, CAPS, BENCH LEDGES

PL1 castor pads PW1-PW4: grind the post/rail bottom seams flush, clamp pads to the flat table (frame inverted), 3 fillet on the inner edges + seal outer edges, grind outer edges flush (det. E).
PL2 caps T1-T4: 3 seal fillet all round, grind flush (det. D).
F3 ledges are welded on the bench BEFORE the box (rail clamped to a straight-edge, stitches from the centre out). A ledge makes the piece position-specific - paint-mark it:
K03 C1 BASE FRONT + F3, top 18 below rail top
K04 C2 BASE REAR + F3, top 18 below rail top
K07 C3 MID REAR + F3, top 12 below rail top
K10 C3 SHELF REAR + F3, top 18 below rail top
Ledge on the face opposite the rear-panel holes (C1: either side); rail top = the face opposite the skirt holes.

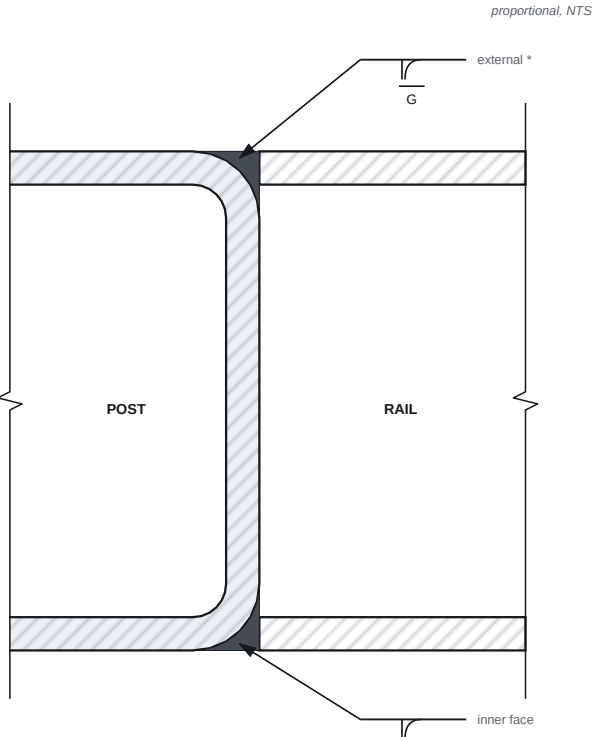
PROJECT	SRA-16 Silent AC Rack - welded 30x30x2.0 frame
TITLE	Box assembly - cross rails, pads, caps, checks
DOC NO.	CP-SRA16-FRM-001 SHEET 3 OF 6 REV A
SCALE	1:10 @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-09-27 DRAWN Claude (AI) for Carbon Project
STATUS	FOR FABRICATION after hold point H1 (AC grille split M3)

A SHS T-JOINT - ELEVATION



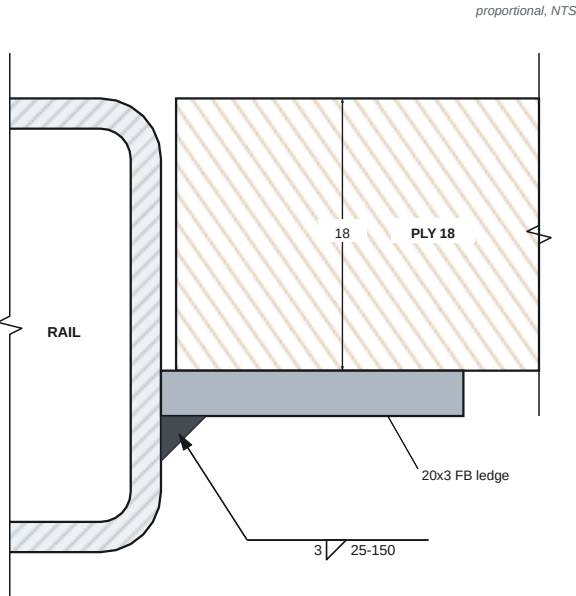
Square-cut end, fit tight (gap 0-1). Weld ALL ROUND: 3 fillet where the rail meets the post face (shown), flare seam where the faces run flush (det. B). Tack 2 per joint first.

B FLUSH FACES - SECTION



Section on the rail centreline. The tube corner radii leave a flare groove where faces run flush: fill it in one pass. Faces marked * carry panels, seals or pads: grind FLUSH (G). Inner faces stay as welded.

C LEDGE ON RAIL - SECTION



Ledge top 18 below the rail top for the floor and shelf, 12 for the partition: set it with an offset of the real ply. Stitch the UNDERSIDE only - the top face is the ply seat.

WELDING PROCEDURE (WPS-lite)

Process	GMAW (MIG), short-circuit transfer
Wire	ER70S-6 (AWS A5.18) ~ ISO 14341-A G 42 4 M21 3Si1, 0.8
Gas	Ar + 15-18% CO2 (e.g. Argoshield Light), 12-15 L/min
Start settings	17-18.5 V, WFS 4.5-6 m/min (~70-100 A)
	stick-out 8-10, push 10-15 deg - prove on a coupon
Fillet	3 leg (2.1 throat) on 2 wall; tacks 6-8 long
Positions	1F/1G, 2F; short 3F uphill OK; no overhead
Preheat / PWHT	none (C350L0, t < 10)
Heat / distortion	balanced order (joint IDs), cool between
Standard	AS/NZS 1554.1 category GP
Inspection	100% visual: no cracks, burn-through, LOF; undercut <= 0.5; no porosity on ground faces
Galvanised stock	grind zinc 25 each side, fume extraction

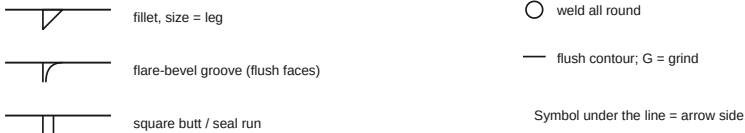
FABRICATION TOLERANCES

Cut length (same mark matched <= 0.5)	+/-0.5
Cut squareness across 30 face	<= 0.5
Rail levels from datum A	+/-1
Outside W x D x H	+/-1.5
Face diagonals (equal)	+/-2
Side frame diagonals (equal)	+/-1.5
Panel faces flat (straight-edge)	<= 2
Twist / pads coplanar	<= 1
Rivnut hole positions	+/-1
Member straightness	L/1000

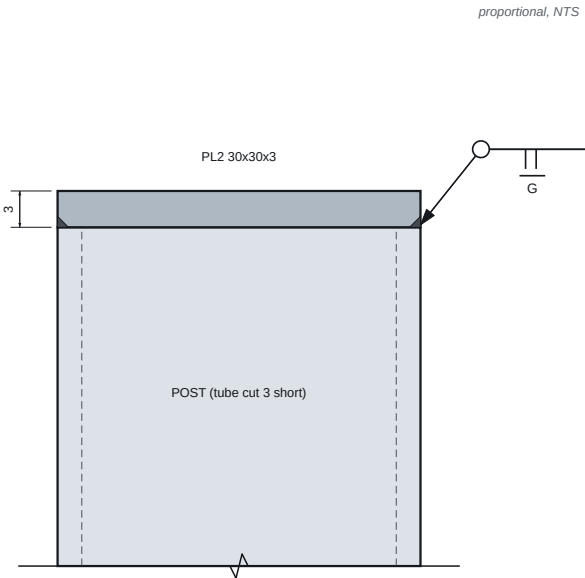
FINISH

Remove spatter; external * faces ground flush (80 grit), edges broken. Blast Sa 2.5 (or chemical pretreat - tubes are perforated by rivnut holes: coater to drain and dry them). Zinc-rich primer + polyester powder, satin black or RAL 7016. Mask the earth point. Fit rivnuts after coating; bond frame to PE (M6, star washer, bare metal) and the 19in strips with 4 mm2 jumpers.

WELD SYMBOLS (AS 1101.3)



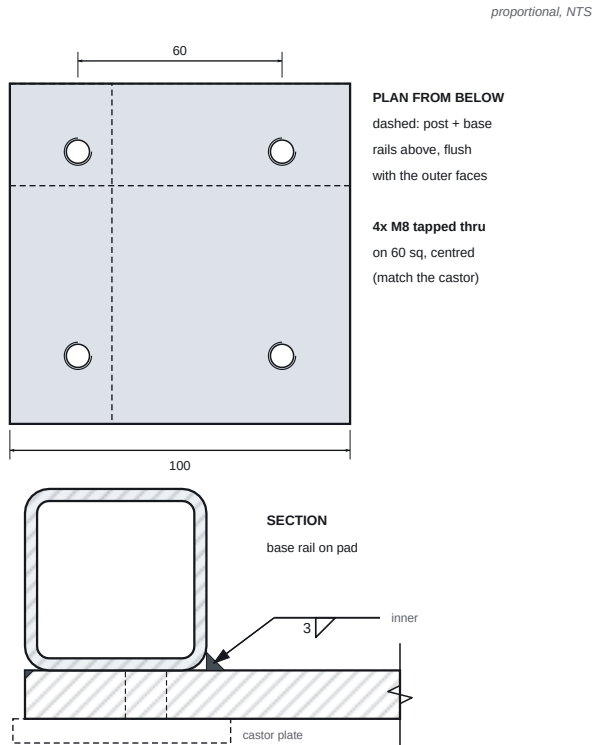
D POST TOP CAP PL2 - ELEVATION



Cap top = frame top (datum A + 1766.2). Top rails butt against the cap edge.

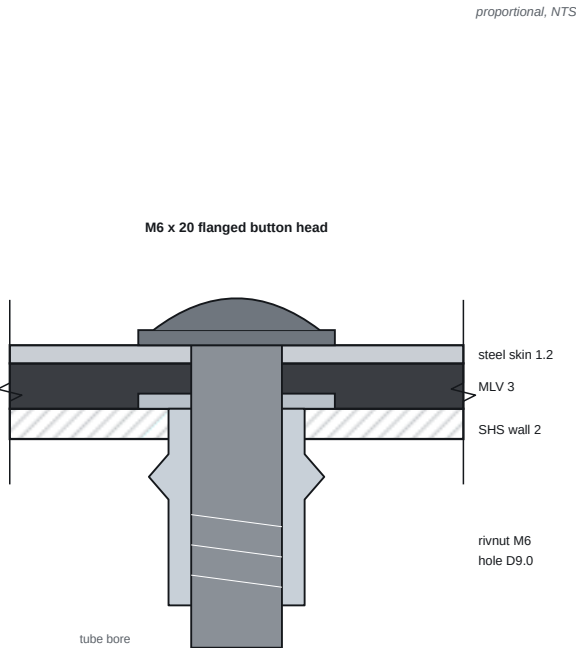
Cap flush with the tube faces all round: seal weld the seam (outside corner) and grind flush on all faces - the top panel and side panels bear here. Weld after the box (step 7).

E CASTOR PAD PL1



Outer edges flush with the frame: seal weld, grind flush. Weld with the four pads clamped to the flat table (coplanar). Castor bolts: plate t + 8 max - must NOT reach the rail walls.

F M6 RIVNUT PANEL FIXING



Drill before welding, deburr. Fit rivnuts (steel, flat head, grip 0.5-3) after coating. M6 x 20: skin + MLV 4.2, the rest into the rivnut (6 min engaged). MLV punched D8 = gasket.

PROJECT	SRA-16 Silent AC Rack - welded 30x30x2.0 frame
TITLE	Weld details, procedure, tolerances, finish
DOC NO.	CP-SRA16-FRM-001 SHEET 4 OF 6 REV A
SCALE	details NTS (proportional) @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-09-27 DRAWN Claude (AI) for Carbon Project
STATUS	FOR FABRICATION after hold point H1 (AC grille split M3)

CUT LIST - CP-SRA16-FRM-001 (all members square cut, deburred)

MARK	QTY	DESCRIPTION	SECTION	CUT	HOLES	KG EA	KG	STAGE	WHERE
P1	2	Post	30x30x2.0 SHS	1763.2	8	2.96	5.92	SA-L/R	post FL; post FR
P2	1	Post	30x30x2.0 SHS	1763.2	15	2.96	2.96	SA-L	post RL
P3	1	Post	30x30x2.0 SHS	1763.2	15	2.96	2.96	SA-R	post RR
S1	4	Side rail	30x30x2.0 SHS	1004.0	9	1.68	6.74	SA-L/R	Base left; Base right; Top left; Top right
S2	4	Side rail	30x30x2.0 SHS	1004.0	5	1.68	6.74	SA-L/R	Mid left; Mid right; Shelf left; Shelf right
C1	1	Cross rail	30x30x2.0 SHS	554.0	3	0.93	0.93	BOX	Base front
C2	2	Cross rail	30x30x2.0 SHS	554.0	5	0.93	1.86	BOX	Base rear; Top rear
C3	3	Cross rail	30x30x2.0 SHS	554.0	3	0.93	2.79	BOX	Mid rear; Shelf rear; Top front
C4	1	Cross rail	30x30x2.0 SHS	554.0	0	0.93	0.93	BOX	Shelf front
U1	4	Rack upright	30x30x2.0 SHS	731.2	7	1.23	4.91	SA-L/R	upright front left; upright front right; upright rear left; upright rear right
RS1	4	Rail spacer (LOOSE - bolted, not welded)	30x30x2.0 SHS	731.2	0	1.23	4.91	LOOSE	spacer front left; spacer front right; spacer rear left; spacer rear right
F1	4	Ledge (flat bar)	20x3 FB	1004.0	0	0.47	1.89	SA-L/R	floor left; floor right; shelf left; shelf right
F2	2	Ledge (flat bar)	20x3 FB	611.0	0	0.29	0.58	SA-L/R	partition left; partition right
F3	4	Ledge (flat bar)	20x3 FB	514.0	0	0.24	0.97	BENCH	floor front; floor rear; partition rear; shelf rear
PL1	4	Castor pad, 4x M8 tapped	PL 100x100x8	-	4 M8	0.63	2.51	FINAL	corner FL; corner FR; corner RL; corner RR
PL2	4	Post top cap	PL 30x30x3	-	-	0.02	0.08	FINAL	top of each post

Weldment 43.1 kg incl. caps and weld metal (0.34 kg); spacers RS1 4.9 kg loose. SHS cut lengths of posts are 3 short of the frame height (cap PL2 on top).

STOCK NESTING (first-fit, kerf 3, 10 trim per bar)

SHS 30x30x2.0 SHS, 6.5 m lengths

bar 1	P1 1763.2 P1 1763.2 P2 1763.2 S1 1004.0	offcut 184
bar 2	P3 1763.2 S1 1004.0 S2 1004.0 S2 1004.0 S2 1004.0 C1 554.0	offcut 139
bar 3	S2 1004.0 S1 1004.0 S1 1004.0 U1 731.2 U1 731.2 U1 731.2 U1 731.2	offcut 532
bar 4	RS1 731.2 RS1 731.2 RS1 731.2 RS1 731.2 C2 554.0 C3 554.0 C4 554.0 C3 554.0 C3 554.0 C2 554.0	offcut 211

FB 20x3, 6.0 m lengths

bar 1	F1 1004.0 F1 1004.0 F1 1004.0 F1 1004.0 F2 611.0 F2 611.0 F3 514.0	offcut 217
bar 2	F3 514.0 F3 514.0 F3 514.0	offcut 4439

FRAME PURCHASE LIST (indicative AUD, Sept 2026 - get quotes)

ITEM	QTY	UNIT	AUD/UNIT	AUD
30x30x2.0 SHS C350L0, 6.5 m length (incl. RS1 spacers)	4	len	52.00	208
Flat bar 20x3, 6.0 m length (ledges)	2	len	22.00	44
Flat bar 100x8 (castor pads, 4 x 100 + saw)	0.45	m	32.00	14
Flat bar 30x3 (post caps)	0.2	m	6.00	1
M6 steel rivnut, flat head, grip 0.5-3 (+10%)	170	ea	0.35	59
M8 x 12 set screw 8.8 + spring washer (castors)	16	ea	0.40	6
Welding wire, gas, discs, primer (share)	1	lot	60.00	60
Blast + zinc primer + powder coat (frame)	1	job	320.00	320

Materials AUD 714 (excl. GST). Castors, panels and rack parts are in bom/BOM.md.

LABOUR ESTIMATE (one-off, manual; tube-laser members save ~6 h)

TASK	HOURS
Cut + deburr 27 SHS + 10 FB pieces, plates	3.5
Drill 152 rivnut holes, tap 16 x M8	3.4
Fit + tack SA-L, SA-R	2.5
Fit + tack box, checks	2.0
Ledges (10), pads, caps	1.5
Weld 8.1 m (repositioning, cooling)	2.4
Grind external faces flush, dress	3.0
Final inspection, measure, record	0.8

Total ~19 h; at AUD 110/h ~ AUD 2,100

SHOP TRAVELLER (initial each box as the batch is done)

MARK	QTY	CUT	DEBURR	DRILL	PAINT MARK	BENCH WELD	FIT + TACK	WELD	QC
P1	2	☐	☐	☐	☐	n/a	☐	☐	☐
P2	1	☐	☐	☐	☐	n/a	☐	☐	☐
P3	1	☐	☐	☐	☐	n/a	☐	☐	☐
S1	4	☐	☐	☐	☐	n/a	☐	☐	☐
S2	4	☐	☐	☐	☐	n/a	☐	☐	☐
C1	1	☐	☐	☐	☐	☐	☐	☐	☐
C2	2	☐	☐	☐	☐	☐	☐	☐	☐
C3	3	☐	☐	☐	☐	☐	☐	☐	☐
C4	1	☐	☐	n/a	☐	n/a	☐	☐	☐
U1	4	☐	☐	☐	☐	n/a	☐	☐	☐
RS1	4	☐	☐	n/a	☐	n/a	n/a	n/a	☐
F1	4	☐	☐	n/a	☐	n/a	☐	☐	☐
F2	2	☐	☐	n/a	☐	n/a	☐	☐	☐
F3	4	☐	☐	n/a	☐	☐	☐	☐	☐
PL1	4	☐	☐	☐	☐	n/a	☐	☐	☐
PL2	4	☐	☐	n/a	☐	n/a	n/a	☐	☐

BENCH WELD = F3 ledges onto C1 / C2 / C3 (sheet 3). QC = length, squareness, holes, mark legible. Keep this sheet with the job.

PROJECT	SRA-16 Silent AC Rack - welded 30x30x2.0 frame
TITLE	Cut list, stock nesting, purchase, labour
DOC NO.	CP-SRA16-FRM-001 SHEET 5 OF 6 REV A
SCALE	- @ A3 UNITS mm LAYOUT v1.2.0
DATE	2026-09-27 DRAWN Claude (AI) for Carbon Project
STATUS	FOR FABRICATION after hold point H1 (AC grille split M3)

DRILLING SCHEDULE - D9.0 holes for M6 steel rivnuts, on the face centreline, positions from each member's DATUM END

MARK	QTY	FACE (position)	USE	DATUM END	POSITIONS (mm from datum end)
P1	2	outer side face	Side panel fixing	bottom end	40.0 280.9 521.8 762.7 1003.5 1244.4 1485.3 1726.2
P2	1	outer side face	Side panel fixing	bottom end	40.0 280.9 521.8 762.7 1003.5 1244.4 1485.3 1726.2
P2	1	rear face	Rear panel fixing	bottom end	160.4 401.3 642.2 883.1 1124.0 1364.9 1605.8
P3	1	outer side face	Side panel fixing	bottom end	40.0 280.9 521.8 762.7 1003.5 1244.4 1485.3 1726.2
P3	1	rear face	Rear panel fixing	bottom end	160.4 401.3 642.2 883.1 1124.0 1364.9 1605.8
S1	4	outer side face	Side panel fixing	front end	40.0 271.0 502.0 733.0 964.0
S1	4	bottom (at base) / top (at top)	Plinth skirt fixing (at base) / Top panel fixing (at top)	front end	155.5 386.5 617.5 848.5
S2	4	outer side face	Side panel fixing	front end	40.0 271.0 502.0 733.0 964.0
C1	1	bottom face	Plinth skirt fixing	left end	100.0 277.0 454.0
C2	2	rear face	Rear panel fixing	left end	40.0 277.0 514.0
C2	2	bottom (at base) / top (at top)	Plinth skirt fixing (at base) / Top panel fixing (at top)	left end	158.5 395.5
C3	3	rear (at mid, shelf) / top (at top)	Rear panel fixing (at mid, shelf) / Top panel fixing (at top)	left end	40.0 277.0 514.0
C4	1	-	no holes	left end	-
U1	4	outer side face	Side panel fixing	bottom end	40.0 257.1 474.1 691.2
U1	4	inner face	19in rail-spacer bolts	bottom end	148.5 365.6 582.7
PL1	4	plate	castor bolts	-	4x M8 tapped through (6.8 drill), 60 square on the pad centre

PIECE DRILLING (1:10 - datum end on the left, one bar per drilled face)

